Jenny Wu

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EDUCATION

Cornell University **Ithaca, NY**
Bachelor of Science in Mechanical Engineering *Aug 2024 – May 2028*
Relevant Coursework: Mechanical Design, Mechanics of Engineering Materials, System Dynamics, Fluid Mechanics, Dynamics, Statics & Mechanics of Solids

EXPERIENCE

Cornell Cup Robotics **Ithaca, NY**
Mechanical Design & Safety Lead *Feb 2025 – Present*
- Designed and prototyped mechanical assemblies in Fusion 360 using iterative testing and design refinement
- Improved fabrication workflows, reducing 3D print time by 50% through optimized tolerancing
- Led onboarding and safety training on drill presses, 3D printers, band saws, and grinders for new members
- Presented engineering solutions and live robotics demonstrations at Governors Cup in Washington, DC

Consigli Construction **Manhattan, NY**
Field Operations Intern *Jun 2026 – Aug 2026*
- Coordinated with cross-functional teams on a 650,000-square-foot office-to-residential conversion project
- Analyzed blueprints and field data to identify MEP conflicts and track demolition and scaffolding progress
- Partnered with engineering and architect teams to improve project execution and field decision-making

PROJECTS

SmartSweep **Ithaca, NY**
Mechanical Synthesis Class | Project Lead *Jan 2026 – May 2026*
- Led development of counter-rotating brush system to remove spotted lanternflies during grape harvesting
- Engineered quick-swap shafts for easy brush replacement, support brackets, and motor/bearing mounts; optimized and validated counter-rotating system to achieve 70% removal effectiveness with 0.02 N·m torque
- Presented product recommendations to industry clients; earned 2nd place for "Outstanding Problem/Solution"

International Space Station (ISS) Mimic **Ithaca, NY**
Cornell Cup Robotics | Mechanical Subteam *Mar 2026 – May 2026*
- Developed an XRP-based platform replicating real-time ISS motion in collaboration with Boeing and Intel
- Designed an 11-tooth pinion and 36-tooth spur gear with a 3.27:1 gear ratio, reducing rotational speed by 69%, and integrated a motor mount and snap-fit bearing interface for controlled 360° center pillar rotation
- Built a detachable 2-axis gantry assembly enabling precise belt-driven movement and repositioning

Submersible **Ithaca, NY**
Cornell Cup Robotics | Mechanical Subteam *Jan 2026 – Mar 2026*
- Developed autonomous submersible for EarthEcho's water-quality initiative, enabling microplastics detection
- Designed syringe-based ballast system using worm gear and rack-and-pinion actuation for buoyancy control
- Modeled and 3D-printed modular electronics trays for integration within a sealed PVC pressure vessel

RC Car **Ithaca, NY**
Cornell Cup Robotics | Mechanical Subteam *Aug 2025 – Dec 2025*
- Built a modular 3D-printed RC platform for camera-based real-time AR mapping and pose tracking
- Developed a modular chassis and camera mount using snap-fit and fastened joints across 10+ components
- Improved steering precision through direct servo integration and optimized 0.30 mm snap-fit tolerances

SKILLS

Software: Fusion 360, Creo Parametric, MATLAB/Simulink, Python, Bluebeam, Procore, Excel
Manufacturing: CNC mills & lathes, drill presses, band saws, grinders, 3D printing, GD&T, DFM
Certifications: OSHA 10-Hour Construction | Red Apron Certified | Procore Certified | First Aid/CPR/AED
Interests: Golf, swimming, painting, photography